

Evaluation Framework for Multimodal Language Models Applied to Geospatial Open Data in Agriculture

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Abstract

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1. 1. Introduction

The rapid development of multimodal Large Language Models (LLMs) has unlocked a huge span of unprecedented applications across diverse scientific and industrial domains. Among these, smart agriculture and precision farming stand out as areas of great potential. The recent emergence of geofoundation models (Jakubik et al. (2023); Corley et al. (2026)) and Earth embeddings (Stewart et al. (2026)), coupled with the swift evolution of generative language models, holds immense promise for Earth observation and agricultural management. This progress has been significantly accelerated by open-source initiatives, which have democratized access to foundational models and driven community-led breakthroughs in applied Artificial Intelligence.

In recent years, a growing body of literature has explored the integration of generative AI into agricultural workflows. Many of these initiatives have materialized as conversational assistants tailored for Earth observation and agronomy (Yuan et al. (2025); Kuckreja et al. (2023)). These studies frequently leverage Retrieval-Augmented Generation (RAG), geo-embeddings, or hybrid approaches combining language modeling with classical computer vision tasks, such as the semantic segmentation of satellite imagery and orthophotos. However, despite these technical advances, existing approaches remain largely fragmented. While the outputs generated by these state-of-the-art tools are increasingly accurate and explore new frontiers in data accessibility, they fundamentally remain generic, failing to provide the highly localized, actionable insights required for precise, parcel-level terrain management.

To bridge this gap, we recently proposed a novel multimodal LLM framework capable of generating highly specific, localized responses regarding designated land regions for agricultural applications (Cañada et al. (2025)). This system achieves its specificity through the retrieval, processing, and fusion of diverse, multi-domain open data sources, effectively functioning as a context-amplified tool for land management.

Following the technical validation of this architecture, a critical challenge arises: the robust and methodical evaluation of its

generated outputs. Since the inception of generative AI, evaluation has remained a primary and open line of research, inherent to any application relying on non-deterministic inference. Unlike traditional machine learning paradigms, where evaluation metrics rely on a finite set of ground-truth labels, the space of correct responses in generative modeling is virtually infinite—a single question can be correctly answered in multiple, equally valid ways depending on the reasoning approach. Numerous evaluation frameworks have been proposed to navigate this complexity. For instance, the LLM-as-a-judge paradigm (Zheng et al. (2023))—which we initially employed in our previous work—is widely adopted due to its simplicity and low operational cost. Nevertheless, this technique is highly vulnerable to technical hallucinations, suffers from self-reference bias, and tends to lose consistency over complex reasoning chains. Similarly, frameworks like RAGAS (Es et al. (2025)) have been developed specifically to evaluate RAG pipelines, and substantial research has been devoted to optimizing automated evaluation metrics.

Despite this rich landscape of evaluation tools, none of the existing frameworks align with the complex requirements of our context-amplified architecture. Specifically, evaluating this system demands more than merely assessing the factual correctness of the final output; it requires interrogating how the model utilized the available external context. A rigorous evaluation must distinguish between a response that is "good but generic" and one that is deeply "specific and context-aware." Furthermore, it must discern whether a high-quality response stems merely from the model's robust foundational pre-training, or if it is the result of successful reasoning over the dynamically retrieved multi-domain data.

To address these unmet needs, this paper proposes a novel, reproducible evaluation methodology tailored for context-amplified multimodal LLMs in agricultural data applications. Our approach introduces a multi-stage, human-in-the-loop (hybrid human-machine) framework designed to mitigate automated biases. The most challenging and innovative aspect of this framework lies in quantifying the marginal contributions of

individual architectural modules. The methodology allows to understand not merely the final quality of a generated response, but exactly how and to what extent each specific module influenced that quality. To achieve this, we design a comprehensive ablation study based on a factorial design, systematically evaluating different configurations by isolating, adding, and removing system modules. The evaluation results are subsequently analyzed through collaborative game theory, providing a rigorous mathematical framework to measure the precise impact and value added by each component.

The entirety of this methodology and its associated tools are open source. Furthermore, to ensure complete reproducibility and to provide a valuable asset for the research community, we have published our complete evaluation dataset on Zenodo as supplementary material. This open dataset comprises 143 detailed evaluations, containing the specific agricultural parcels, the exact external context retrieved, and the corresponding responses generated by the model.

The remainder of this article is structured as follows: Section 2 details the architecture of the conversational system, provides a functional overview, and extensively outlines the evaluation methodology, including parcel selection, human-in-the-loop judge guidelines, evaluation stages, utilized metrics, and the design of the ablation study. Section 3 presents and thoroughly discusses the results derived from the evaluation and the game theory analysis. Finally, Section 4 summarizes the main conclusions drawn from this work.

2. Materials and methods

2.1. System architecture

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The proposed system is designed to combine several interconnected modules, which integrate heterogeneous data sources. The overall architecture, included in Fig. 3, is composed of two main layers: user interface and backend processing, which includes the LLM reasoning. The aim of the data flow is to provide the final LLM with the most rich context regarding the area of interest (a specific parcel) in order for it to give the best possible answer to the user's query.

The backend consists of five different blocks, which form a data stream from the user query to the final LLM, and then back to the user with the answer: (i) plot info retrieval, which is responsible for fetching, from the SIGPAC registry, both plot data-as slope, elevation, area, perimeter or land use, and the geo-referenced geometry. Once known the area of interest, the (ii) context retrieval block performs two parallel tasks, each leveraging different open-source data sources to enrich the context: (a) an orthophoto of the plot, obtained from aerial imagery, is processed through a multimodal LLM along with the plot data, generating a detailed description of the terrain. In addition, (b) satellite multispectral data, specifically from Sentinel-2 (Copernicus mission, European Space Agency) are retrieved. After spatial and temporal processing combining multiple spectral bands, an array of vegetation and water indices that reflect biophysical conditions of

the land are obtained. In parallel, (iii) the system queries a document repository through a retrieval-augmented generation mechanism, which provides relevant textual knowledge such as agronomic guidelines, regulations, or best practices. All these multi-domain data are then fused through (iv) an aggregator, which converts all data to natural language, and prepares the information into an encapsulated prompt that provides the (v) final LLM with the overall context of the plot and the query. Several open-source models were evaluated, including Qwen3-32B, Llama-3.3-70B or Mistral-Small-3.2-24B, being Qwen the one chosen for the final implementation due to its superior performance. Regarding the LLMs framework, the model inference is managed with the inference engine vLLM [9], orchestrated through the liteLLM proxy [10], which agglutinates all models in one port. Besides from the multimodal and final LLMs, there are also embeddings, reranker, and tool triage models working together. The infrastructure runs in a 8x80GB cluster of NVIDIA GPUs.

2.2. Evaluation framework

El principal problema que se plantea resolver, además de obtener una evaluación sistemática, rigurosa y reproducible, es el de ser capaces de evaluar no solo si la respuesta es buena, sino también en qué medida esa calidad es gracias a los módulos de su arquitectura.

Internal Knowledge over-reliance → sesgo de conocimiento interno

Therefore, **an experimental evaluation methodology has been designed**, which is detailed in this section. This approach, designed to optimize the consumption of operational resources while maintaining statistical robustness, consists of two main stages: evaluation of the complete system and analysis of marginal contributions through an ablation study.

The main challenge of this experimental methodology, and what differentiates it from other existing approaches, is that it must be able to address the evaluation of geospatial context usage in response generation, not merely the quality of the responses themselves. A response that is correct but generic, or even seemingly specific but generated from the model's base knowledge, is not equivalent to a *personalized* response adapted to terrain characteristics. Therefore, multiple stages are combined, different configurations are evaluated, and specific metrics are proposed.

2.3. Parcel Selection and Heterogeneity

A total of 15 parcels were selected, categorized into four different Land Use (LU) types. The distribution of evaluations is weighted according to the operational utility of the tool for each functional land use, as shown in ??.

To ensure an unbiased evaluation, parcels were selected to represent a high degree of heterogeneity in their physical characteristics, including variable slopes (0.5% to 21.6%), altitudes, irregular shapes, and diversity in vegetation health (NDVI ranges). The parcels are shown in 3.

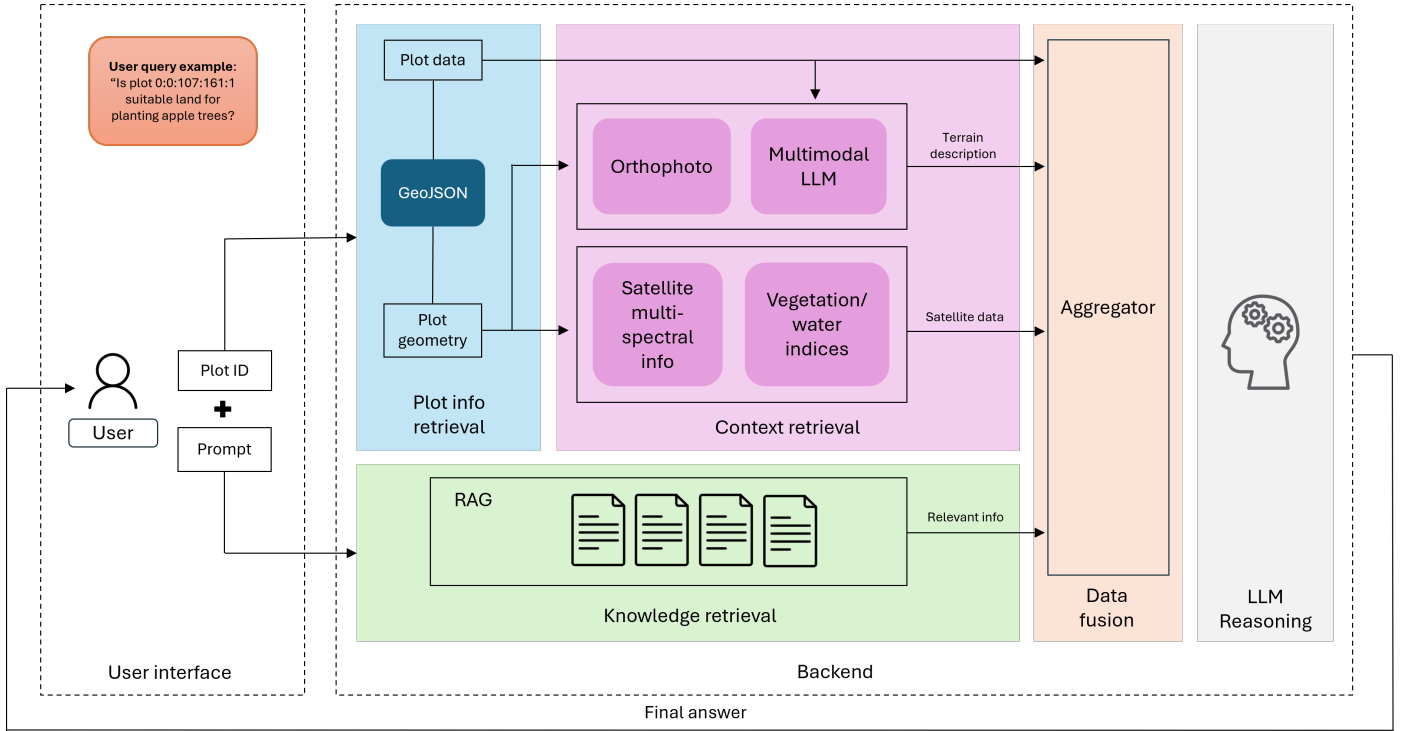


Figure 1:

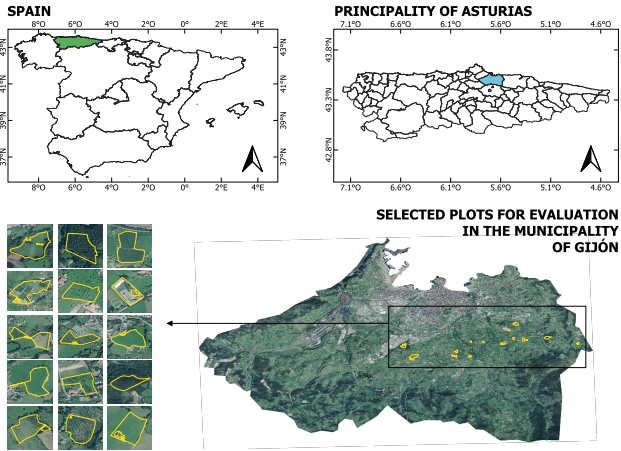


Figure 2:

2.4. Phase I: Broad Performance Assessment

This phase evaluates the overall system capability across diverse scenarios. A total of 71 queries were designed, distributed as follows:

- **Productive Parcels (FR and PA):** 5 queries per parcel, focused on irrigation, fertilization, pest management, and investment ROI.
- **Non-Productive Parcels (UN and FO):** 4 queries per parcel, consisting of 2 functional queries and 2 *Negative Testing Queries* (scenarios where the system must identify unsuitability).

2.5. Phase II: Ablation Study (Architecture Validation)

To quantify the impact and marginal contribution of each architectural module, an ablation study was conducted using a **factorial design**. Ablation studies consist of the individual analysis of different subsets (configurations) of a system, obtained through systematic activation and deactivation of its components. In this case, with three modules (Orthophoto, Satellite, and RAG, previously shown in ??), the factorial design of experiments yields $2^3 = 8$ possible configurations.

- **Configurations:** 8 modular combinations + 1 Zero-Shot baseline (the LLM without parcel context) = 9 total configurations, listed in 1.
- **Sample subset:** 1 representative parcel per Land Use category.
- **Query volume:** 2 specific queries per parcel across the 9 configurations = 72 queries.

2.6. Evaluation Volume Summary

The combined results provide a total set of 143 evaluations (3). In Appendix A, a link to the dataset created for this evaluation is included, where all queries, responses, and context can be consulted.

2.7. Evaluation Subjects and Bias Mitigation

To ensure balanced validation, mitigate individual biases, and capture a wide range of perspectives, the evaluation process is carried out independently by **four differentiated evaluators**:

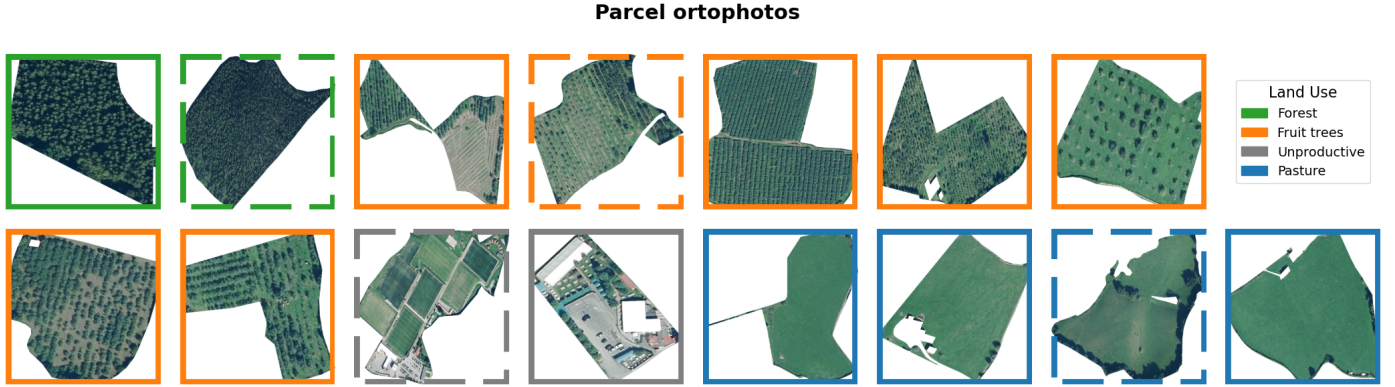


Figure 3: Selection of parcels for tool evaluation, grouped by Land Use. The number of parcels per LU depends on their functional utility in agriculture. Dashed borders correspond to parcels used in ablation study.

Table 1: Configuraciones modulares del estudio de ablaci3n. PD means Plot Data. Active tool is always enabled because it is responsible for identifying parcel IDs as so.

Modules	RAG	ORTHO	SAT	PD	Active Tool	Code
Baseline	0	0	0	1	1	0
B + SAT	0	0	1	1	1	1
B + ORTHO	0	1	0	1	1	2
B + ORTHO + SAT	0	1	1	1	1	3
B + RAG	1	0	0	1	1	4
B + RAG + SAT	1	0	1	1	1	5
B + RAG + ORTHO	1	1	0	1	1	6
Complete	1	1	1	1	1	7
ZERO SHOT	0	0	0	0	1	ZS

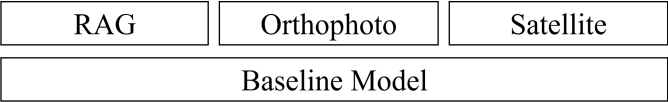


Figure 4:

- **Human Evaluators (2):**
 - *Expert Evaluator:* PhD in Biology.
 - *Control Evaluator:* Non-specialist profile.
- **LLM-as-a-judge (2):**
 - *Proprietary Model (Gemini):* Gemini3.1 Flash Lite
 - *Open Source Model (Qwen):* Qwen3.5-397B-A17B-Instruct

2.8. Multidimensional Metrics Framework

Response quality is measured according to a multidimensional scale consisting of five metrics, divided into those scored considering context and those that are not, as shown in 4. Scores are given on a *Likert* scale (integer values between 1 and 5).

2.9. Preliminary Results and Discussion

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This section presents some preliminary evaluation results. It should be noted that the graphs included here are only an initial overview that sheds light on the system evaluation; more advanced statistical analyses are reserved for a future scientific publication. First, 5 contains the average scores of each of the 4 evaluators across the 5 metrics, grouped by configuration. From this graph, numerous observations can be made, with the most notable being those related to evaluator correlation, and especially regarding the marginal contributions of the modules. Recalling 1, we see that configurations 2 and 3, as well as 6 and 7, include the orthophoto module; and in all metrics, a higher average score is observed in these cases, allowing us to deduce that the marginal contribution of the orthophoto module is significant.

Similarly, the trend across all metrics is ascending along the x-axis (values are lower in left-side configurations), showing how, for example, system utility in the *Zero Shot* configuration is below 1/5 for most evaluators. Since this configuration corresponds to a standard LLM model without context augmentation, and with configurations progressively ordered by active modules, it can be concluded that **responses, as expected, improve as more geospatial terrain information is added**. This result consequently verifies the initial hypothesis and validates the proposed architecture.

More generally, ?? represents the average metric values from

Table 2: Distribution of parcels, queries, and configurations across methodology stages for each land use category.

Land Use	Broad			Ablation				Total
	No. parcels	Queries/parcel	No. queries	No. parcels	Queries/parcel	No. configs	No. queries	
Fruit trees (FR)	7	5	35	1	2	9	18	—
Pasture (PA)	4	5	20	1	2	9	18	—
Forest (FO)	2	4	8	1	2	9	18	—
Unproductive (UN)	2	4	8	1	2	9	18	—
Total			71				72	143

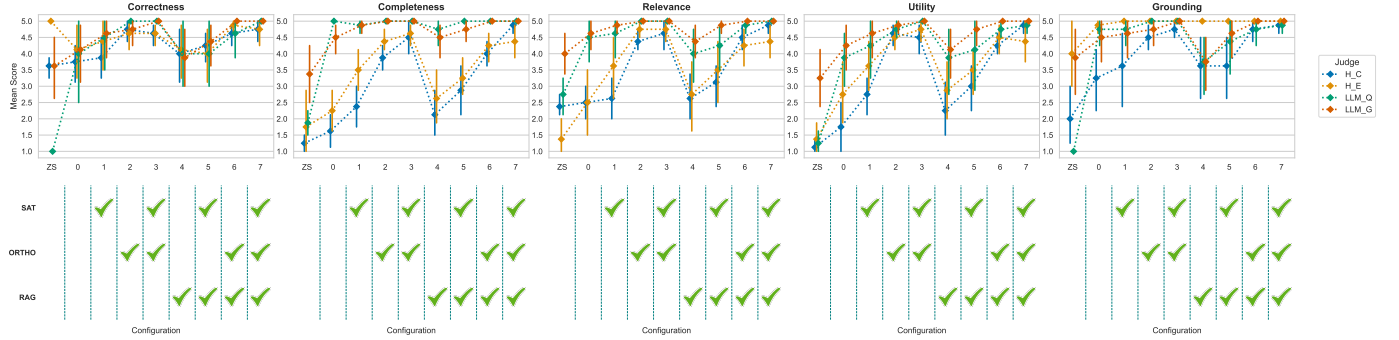
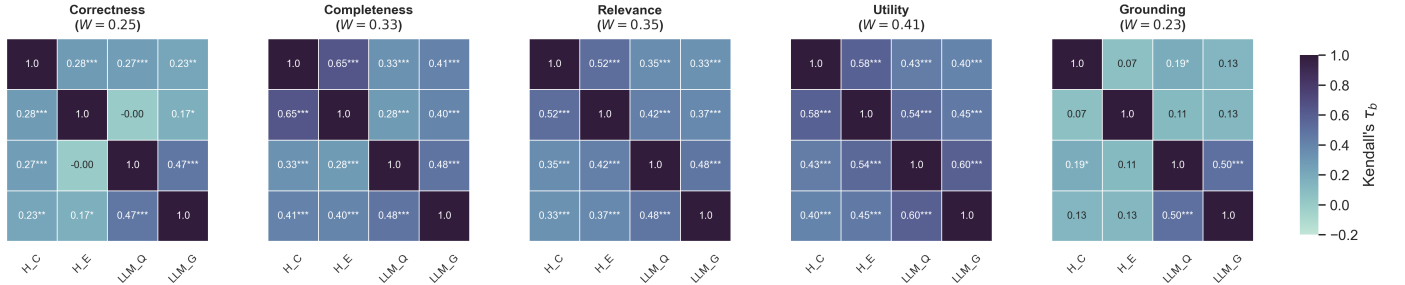
Figure 5: Preliminary evaluation results from the *Ablation Study* phase for the 4 evaluators and 5 metrics grouped by configuration. Differences can be observed based on active modules.

Figure 6:

Table 3: Summary of total query volume in the evaluation.

Evaluation Phase	Methodology	Num. of queries
Broad Assessment	Sequential	71
Ablation Study	Simultaneous	72
Total		143

the *Broad* stage evaluations, i.e., the complete system. For all evaluators, the resulting metrics average values very close to 5/5, implying that the system has responded with a high degree of success to the evaluated queries. These preliminary results serve as evidence that the proposed system is capable of generating high-quality, complete, useful, and generally successful responses in the broad sense.

Shapley Values

Table 4: Multidimensional evaluation metrics framework. Note: The *Grounding* metric measures the degree to which responses are derived from the obtained context, rather than generic knowledge, making it the only metric requiring context for evaluation. The *Correctness* metric does not apply to the human control evaluator.

Metric	Type
Correctness	Without Context
Completeness	Without Context
Relevance	Without Context
Utility	Without Context
Grounding	With Context

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S)) \quad (1)$$

N is the set of all players: $N = \{M_S, M_O, M_R\}$ and S is any

Table 5: .

Metric	H_C	H_E	LLM_Q	LLM_G
Correctness	4.690	4.704	4.817	4.746
Completeness	4.803	4.817	4.944	4.958
Relevance	4.549	4.817	4.831	4.930
Utility	4.704	4.761	4.859	4.944
Grounding	4.606	4.901	4.296	4.549

subset or *coalition*.

$$\begin{aligned}\phi_S(v) = & \frac{1}{3}[v(\{M_S\}) - v(\emptyset)] \\ & + \frac{1}{6}[v(\{M_S, M_O\}) - v(\{M_O\})] \\ & + \frac{1}{6}[v(\{M_S, M_R\}) - v(\{M_R\})] \\ & + \frac{1}{3}[v(\{M_S, M_O, M_R\}) - v(\{M_O, M_R\})]\end{aligned}\quad (2)$$

$$\begin{aligned}\phi_O(v) = & \frac{1}{3}[v(\{M_O\}) - v(\emptyset)] \\ & + \frac{1}{6}[v(\{M_S, M_O\}) - v(\{M_S\})] \\ & + \frac{1}{6}[v(\{M_O, M_R\}) - v(\{M_R\})] \\ & + \frac{1}{3}[v(\{M_S, M_O, M_R\}) - v(\{M_S, M_R\})]\end{aligned}\quad (3)$$

$$\begin{aligned}\phi_R(v) = & \frac{1}{3}[v(\{M_R\}) - v(\emptyset)] \\ & + \frac{1}{6}[v(\{M_S, M_R\}) - v(\{M_S\})] \\ & + \frac{1}{6}[v(\{M_O, M_R\}) - v(\{M_O\})] \\ & + \frac{1}{3}[v(\{M_S, M_O, M_R\}) - v(\{M_S, M_O\})]\end{aligned}\quad (4)$$

$$\begin{aligned}\phi_S(v) = & \frac{1}{3}[v(C_1) - v(C_0)] + \frac{1}{6}[v(C_3) - v(C_2)] \\ & + \frac{1}{6}[v(C_5) - v(C_4)] + \frac{1}{3}[v(C_7) - v(C_6)]\end{aligned}\quad (5)$$

$$\begin{aligned}\phi_O(v) = & \frac{1}{3}[v(C_2) - v(C_0)] + \frac{1}{6}[v(C_3) - v(C_1)] \\ & + \frac{1}{6}[v(C_6) - v(C_4)] + \frac{1}{3}[v(C_7) - v(C_5)]\end{aligned}\quad (6)$$

$$\begin{aligned}\phi_R(v) = & \frac{1}{3}[v(C_4) - v(C_0)] + \frac{1}{6}[v(C_5) - v(C_1)] \\ & + \frac{1}{6}[v(C_6) - v(C_2)] + \frac{1}{3}[v(C_7) - v(C_3)]\end{aligned}\quad (7)$$

Meter gráficas solo con valores del promedio de los 4 jueces para no crear ruido visual

3. 3. Results and Discussion

Referencia a zenodo con dataset anonimizado

4. 4. Conclusions

Acknowledgements

Thanks to ...

Appendix A. Appendix A: Shapley values

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Table 6: Shapley

Judge	Metric	Module	Shapley Value	Relative Contribution (%)	Global Delta
H_C	Correctness	R	0.167	16.67%	1.000
		O	0.729	72.92%	
		S	0.104	10.42%	
	Completeness	R	0.396	12.18%	3.250
		O	2.083	64.10%	
		S	0.771	23.72%	
	Relevance	R	0.229	9.65%	2.375
		O	1.854	78.07%	
		S	0.292	12.28%	
	Utility	R	0.271	8.67%	3.125
		O	2.208	70.67%	
		S	0.646	20.67%	
	Grounding	R	0.208	12.82%	1.625
		O	1.208	74.36%	
		S	0.208	12.82%	
H_E	Correctness	R	0.021	3.33%	0.625
		O	0.583	93.33%	
		S	0.021	3.33%	
	Completeness	R	-0.021	-0.98%	2.125
		O	1.541	72.55%	
		S	0.604	28.43%	
	Relevance	R	-0.146	-7.78%	1.875
		O	1.479	78.89%	
		S	0.542	28.89%	
	Utility	R	-0.104	-6.41%	1.625
		O	1.333	82.05%	
		S	0.396	24.36%	
	Grounding	R	0.042	33.33%	0.125
		O	0.042	33.33%	
		S	0.042	33.33%	
LLM_Q	Correctness	R	-0.146	-14.58%	1.000
		O	0.854	85.42%	
		S	0.292	29.17%	
	Completeness	R	-0.062	0.00%	0.000
		O	0.062	0.00%	
		S	0.000	0.00%	
	Relevance	R	-0.250	-50.00%	0.500
		O	0.625	125.00%	
		S	0.125	25.00%	
	Utility	R	-0.062	-6.25%	1.000
		O	0.812	81.25%	
		S	0.250	25.00%	
	Grounding	R	-0.479	-383.33%	0.125
		O	0.458	366.67%	
		S	0.146	116.67%	
LLM_G	Correctness	R	-0.083	-9.52%	0.875
		O	0.667	76.19%	
		S	0.292	33.33%	
	Completeness	R	-0.021	-4.17%	0.500
		O	0.354	70.83%	
		S	0.167	33.33%	
	Relevance	R	-0.083	-22.22%	0.375
		O	0.292	77.78%	
		S	0.167	44.44%	
	Utility	R	0.000	0.00%	0.750
		O	0.500	66.67%	
		S	0.250	33.33%	
	Grounding	R	-0.208	-41.67%	0.500
		O	0.479	95.83%	
		S	0.229	45.83%	

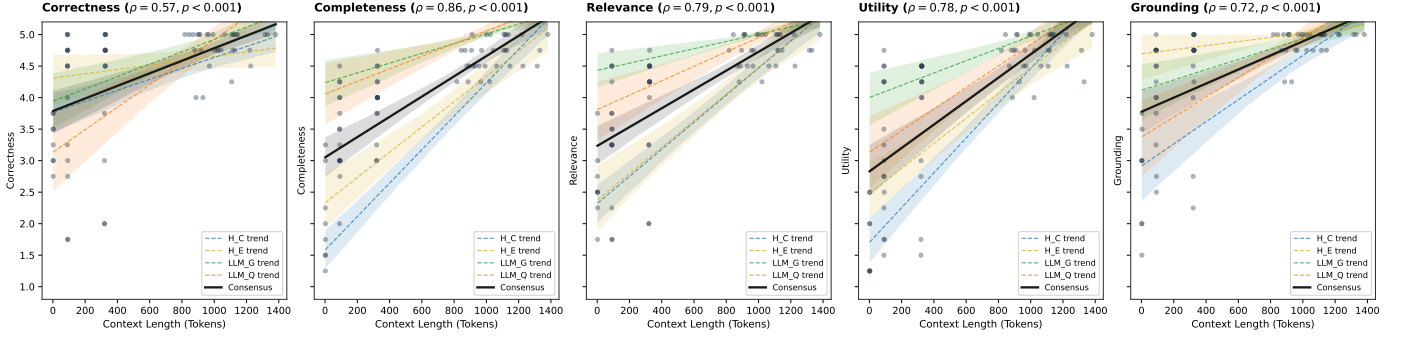


Figure 7:

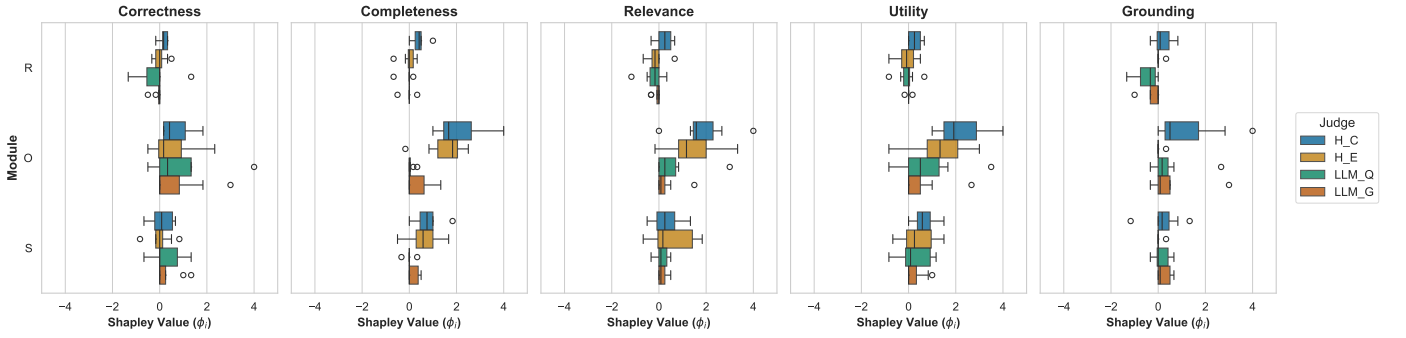


Figure 8:

Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I., 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. URL: <http://arxiv.org/abs/2306.05685>, doi:10.48550/arXiv.2306.05685. arXiv:2306.05685 [cs].

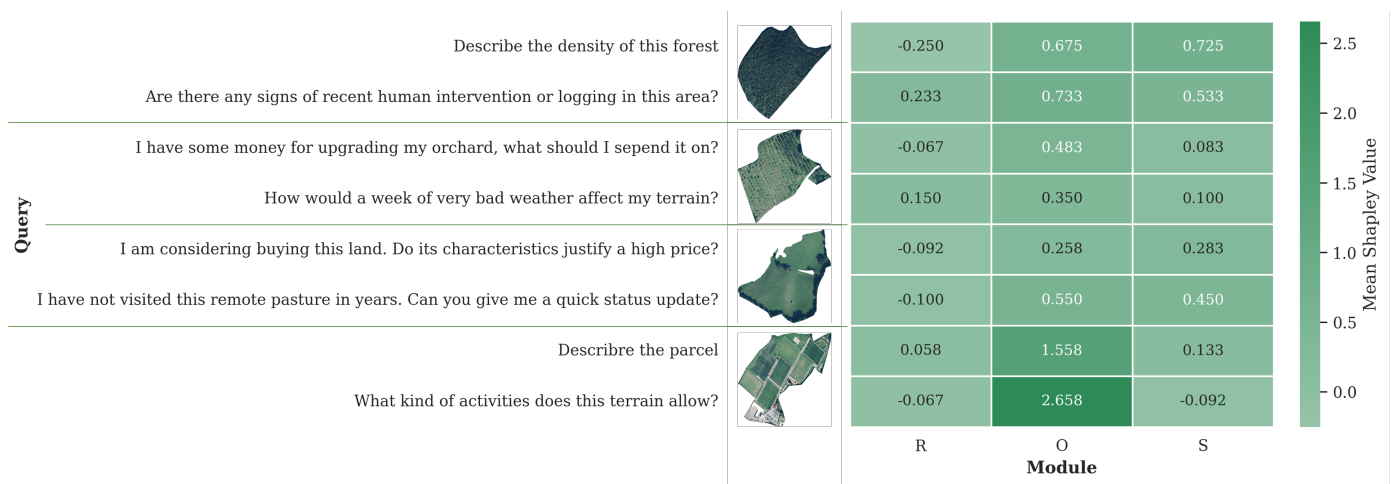


Figure 9: